

EX-17

USING FUNCTIONS

PROBLEM 1

```
mysql> CREATE TABLE customer (  
->     customer_id INT,  
->     customer_name VARCHAR(30),  
->     credit_limit DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.12 sec)  
  
mysql>  
mysql> INSERT INTO customer VALUES  
-> (1, 'Arjun', 60000),  
-> (2, 'Kavin', 30000),  
-> (3, 'Ravi', 5000);  
Query OK, 3 rows affected (0.04 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql>  
mysql> DELIMITER //  
mysql>  
mysql> CREATE FUNCTION customer_level(climit DECIMAL(10,2))  
-> RETURNS VARCHAR(10)  
-> DETERMINISTIC  
-> BEGIN  
->     DECLARE level_name VARCHAR(10);  
->  
->     IF climit > 50000 THEN  
->         SET level_name = 'PLATINUM';  
->     ELSEIF climit >= 10000 AND climit <= 50000 THEN  
->         SET level_name = 'GOLD';  
->     ELSE  
->         SET level_name = 'SILVER';  
->     END IF;  
->  
->     RETURN level_name;  
-> END //  
Query OK, 0 rows affected (0.06 sec)  
  
mysql>  
mysql> DELIMITER ;  
mysql>  
mysql> SELECT customer_id, customer_name, credit_limit,  
->         customer_level(credit_limit) AS customer_level  
-> FROM customer;
```

customer_id	customer_name	credit_limit	customer_level
1	Arjun	60000.00	PLATINUM
2	Kavin	30000.00	GOLD
3	Ravi	5000.00	SILVER

3 rows in set (0.03 sec)

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PROBLEM 2

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE factorial(IN n INT, OUT result BIGINT)
-> BEGIN
->     IF n <= 1 THEN
->         SET result = 1;
->     ELSE
->         CALL factorial(n - 1, result);
->         SET result = n * result;
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SET @fact = 0;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CALL factorial(5, @fact);
ERROR 1456 (HY000): Recursive limit 0 (as set by the max_sp_recursion_depth variable) was exceeded for routine factorial
mysql>
mysql> SELECT @fact AS factorial;
+-----+
| factorial |
+-----+
|          0 |
+-----+
1 row in set (0.00 sec)

mysql>
```